# Tribhuvan University

## Faculty of Humanities and Social Sciences

# WEB DEVELOPER AT DANPHELINK

## INTERNSHIP REPORT

**Submitted to**

Department of Computer Application

GP Koirala Memorial College

Siphal, Kathmandu

**Under the supervision of**

Mr. Shyam Sundar khatiwada

***In partial fulfillment of the requirements for the Bachelor's in Computer Application***


Submitted by

Biplov Karki

6-2-703-6-2020


16th March, 2

## Tribhuvan University

### Faculty of Humanities and Social Sciences

### G.P. Koirala Memorial College

Sifal, Kathmandu

Bachelor of Computer Applications (BCA)

# CERTIFICATE OF COMPLETION

I hereby recommend that this internship report is prepared under my supervision by **Biplov Karki** entitled **“Web Developer”** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

…………………………………………

Signature

Shyam Sunder Khatiwada

Supervisor

GP Koirala Memorial College

**Tribhuvan University**

**Faculty of Humanities and Social Sciences**

**G.P. Koirala Memorial College**

Sifal, Kathmandu

Bachelor of Computer Applications (BCA)

# SUPERVISOR'S RECOMMENDATION

I hereby recommend that this internship report is prepared under my supervision by **Biplov Karki** entitled **"Web Developer"** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

…………………………………………

Signature

Shyam Sunder Khatiwada

Supervisor

GP Koirala Memorial College

**Tribhuvan University**

**Faculty of Humanities and Social Sciences**

**G.P. Koirala Memorial College**

Sifal, Kathmandu

Bachelor of Computer Applications (BCA)

# LETTER OF APPROVAL

This is to certify that this internship report prepared by **BIPLOV KARKI** entitled **"WEB DEVELOPER"** in partial fulfillment of the requirements for the degree of Bachelor's in Computer Application has been well studied. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

| | |
|---|---|
| ……………………………………<br>Shyam Sunder Khatiwada<br>Supervisor<br>GPKMC | ………………………………...<br>Shiva Adhikari<br>Campus Chief<br>GPKMC |
| ……………………………………<br>Sujan Shrestha<br>Internship Coordinator<br>GPKMC | ………………………………...<br>[Name]<br>External Supervisor<br>Designation |

# MENTOR'S RECOMENDATION

DanpheLink
YOUR PARTNER IN TECH EVOLUTION

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2nd Floor, Rem Work Building,
Kamalpokhari 01, Kathmandu, Nepal.
www.danphelink.com.np

## Mentor Recommendation

To whom it may concern

I am pleased to recommend Mr. Biplov, who completed his internship at DanpheLink Pvt. Ltd. as a Frontend Developer Intern under my mentorship.

He demonstrated strong skills in frontend development using React.js and Next.js, along with a solid understanding of UI/UX principles. Biplov effectively used Figma for designing intuitive, user-friendly interfaces and translated those designs into responsive web components. He was quick to adapt, delivered quality work on time, and showed a strong eagerness to learn.

I recommend him for any opportunity in frontend development and UI/UX design.

Mr. Shankar Kumar
Yadav Internship Mentor
Danphelink Pvt.Ltd.
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# DECLARATION

I, Mr. Biplov Karki hereby declare that the report of internship program titled “Web Development” is uniquely prepared by me.

I confirm that, the report is only prepared for my academic requirement not for other purpose.

It might be with the interest of opposite party of the corporation. I also assure that this report is not submitted anywhere of Nepal before me.

# ACKNOWLEDGMENT

I would like to express my sincere gratitude to **DanpheLink** for providing me with the opportunity to undertake this internship, which has been a valuable learning experience in my academic and professional journey.

My heartfelt appreciation goes to my **internship supervisor and mentors** at DanpheLink for their guidance, support, and insightful feedback throughout the internship period. Their expertise in the **FrontEnd Development** and encouragement have significantly contributed to my learning and skill development.

I am also grateful to **Mr. Shyam Sunder Khatiwada**, our project supervisor, for his constant motivation and guidance. Special thanks to **Mr. Shiva Adhikari**, the **Campus Chief**, for facilitating this internship opportunity and supporting students in their professional growth.

I extend my appreciation to **Mr. Sujan Shrestha**, the **BCA Program Coordinator**, for his continuous support in shaping our academic journey and ensuring the successful completion of this internship.

A sincere thank you to **GP Koirala Memorial College** and its faculty members for their encouragement and the knowledge they have imparted, which played a crucial role in my performance during the internship.

Finally, I extend my gratitude to **Tribhuvan University** for including internship programs as part of the BCA curriculum, enabling students to gain practical industry experience and bridge the gap between academic knowledge and real-world application.

This internship has been an enriching experience, helping me strengthen my skills in frontend development while gaining exposure to industry standards, teamwork, and real-world project development.

Biplov Karki

Registration No.: 6-2-703-6-2020

Date: March 16, 2025

# ABSTRACT

This report presents an overview of my internship experience at DanpheLink, where I worked as a Frontend Development Intern with a primary focus on frontend development. The goal of this internship was to bridge the gap between academic learning and real-world industry practices by working on practical projects using modern web technologies.

During this period, I gained hands-on experience with React.js, Next.js, TypeScript, Tailwind CSS, and other frontend tools, refining my skills in UI/UX design, state management, and API integration. Collaborating within an agile team, I developed a deeper understanding of frontend development workflows, debugging techniques, performance optimization, and industry best practices.

This report details the tasks I handled, the challenges I faced, and the solutions I implemented. It also highlights essential takeaways such as teamwork, time management, and effective communication. Additionally, it reflects on how this internship contributed to my growth as a developer and prepared me for future opportunities in the software industry.

Overall, my internship at DanpheLink was a highly valuable experience, allowing me to apply theoretical knowledge in real-world scenarios while strengthening both my technical and professional skills. The insights gained will serve as a strong foundation for my future career in frontend web development.

**Keywords:** Internship, Frontend Development, React.js, Next.js, Web Technologies, Professional Growth

# LIST OF ABBREVIATIONS

| | |
|---|---|
| API | Application Programming Interface |
| CI/CD | Continuous Integration / Continuous Deployment |
| CMS | Content Management System |
| CRUD | Create, Read, Update, Delete |
| IDE | Integrated Development Environment |
| JSON | JavaScript Object Notation |
| JSX | JavaScript XML |
| SEO | Search Engine Optimization |
| SPA | Single Page Application |
| TS | TypeScript |
| TSX | TypeScript XML |
| UI | User Interface |
| UX | User Experience |
| VFX | Visual Effects |

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# CHAPTER 1: INTRODUCTION

## 1.1 Introduction

The purpose of my internship at DanpheLink Pvt. Ltd. was to gain hands-on experience in web development, particularly focusing on frontend development, UI/UX design, and website responsiveness. As someone passionate about building user-friendly interfaces and optimizing digital experiences, I saw this opportunity as a way to enhance my skills while contributing to a meaningful project.

**DanpheLink Academy** (danphelink.edu.np), being an institution dedicated to IT education, required a more streamlined, visually appealing, and interactive web platform to cater to the needs of its students, instructors, and prospective learners. A well-structured website plays a crucial role in establishing credibility, improving accessibility, and ensuring a seamless learning experience. With the growing demand for digital education, optimizing the academy's online presence was essential in creating a platform that is both informative and engaging for users.

My role during the internship involved analyzing the existing website, identifying areas for improvement, and implementing changes to enhance both functionality and aesthetics. I worked on UI design refinements, responsive layout optimization, and form-handling enhancements, ensuring that the platform was accessible across all devices. I collaborated closely with the development team to redesign core sections, such as the homepage, course pages, contact page, and enrollment forms, with the goal of improving navigation, readability, and user engagement.

Additionally, I conducted research on modern web technologies, particularly TypeScript, to improve code maintainability, scalability, and overall performance. By integrating TypeScript into the project, I aimed to make the website more robust, reducing potential errors, and ensuring a smoother development process in the long run.

This internship not only allowed me to apply my technical knowledge in a real-world setting but also gave me valuable insights into the importance of UI/UX design principles, cross-platform compatibility, and structured development practices. Through hands-on collaboration and problem-solving, I was able to enhance my skills while contributing to DanpheLink Academy's goal of providing a better digital experience for its users.

## 1.2 Statement of Problem

During my internship at DanpheLink Pvt. Ltd., I encountered several challenges related to frontend development and backend integration as the company worked toward creating a more interactive and user-friendly digital platform. While the goal was to improve the overall website experience, it became clear that there were several areas in need of significant improvement, particularly in terms of responsiveness, usability, and performance.

The primary issue was the lack of mobile optimization, which led to misaligned elements and poor navigation, making it difficult for users to interact with the platform on different devices. Furthermore, the website's reliance on outdated web technologies resulted in performance bottlenecks and limited maintainability. The code base was complex and poorly structured, which posed challenges for future scalability and the smooth integration of new features. The inefficient integration between the frontend and backend also caused slow load times and issues with dynamic data rendering, further affecting user experience.

Additionally, the interactive components, such as forms, buttons, and navigation, were not designed with a user-centric approach. They lacked intuitive design and responsiveness, creating barriers for users trying to complete tasks efficiently. These issues, combined with user feedback and performance data, highlighted the need for a more structured and responsive design framework to enhance both accessibility and user engagement.

The key challenges identified were:

- **Underutilization of Modern Web Technologies:** The website's reliance on outdated technologies hampered performance and responsiveness, especially on mobile devices.
- **Poor Mobile Optimization:** Layouts were inconsistent across different screen sizes, which detracted from the overall user experience.
- **Non-Intuitive Interactive Components:** Forms, buttons, and navigation lacked user-friendly features, making it difficult for users to complete tasks smoothly.
- **Unorganized and Difficult-to-Maintain Codebase:** The existing code structure made it challenging to scale the website and efficiently implement new features.
- **Accessibility and Usability Issues:** Performance analysis and user feedback indicated difficulties in accessibility, creating friction for users interacting with the platform.

Addressing these challenges was crucial to building a more seamless, engaging, and accessible experience for users, particularly students and learners who rely on the platform for their educational needs.

## 1.3 Objectives

The main objectives of the internship include:

- To Optimize Website Responsiveness
- To Adopt Modern Technologies
- To Improve User Experience (UX)
- To Streamline Codebase

## 1.4. Scope and Limitations

The scope of Internship are:

- Internships provide valuable learning opportunities for students or individuals looking to gain practical experience in a particular field.
- Internships offer a chance to develop and enhance skills related to the chosen field, such as communication, problem-solving, technical abilities, and teamwork.
- Internships provide an opportunity to network with professionals in the industry, including supervisors, mentors, and colleagues.
- Internships add credibility to a resume and demonstrate practical experience in a specific field.
- Internships allow individuals to explore different career paths within their field of interest.

The limitations of Internship are:

- Internships typically have a fixed duration, which may range from a few weeks to several months.
- Limited supervision can impact the quality of learning and support available to interns.
- Interns may be assigned tasks or projects with limited scope and complexity, depending on their level of experience and the organization's needs.
- Financial constraints may restrict the ability of some individuals to participate in unpaid internships.
- Internships do not guarantee future employment with the organization.

## 1.5. Report Organization

The report is based on five chapters.

**Introduction**

Provides an overview of the internship, including the problem statement, objectives, scope, and limitations of the internship.

**Organization Overview**

Details about DanpheLink, including its domain and the department where the internship was conducted.

**Background Study and Literature Review**

Reviews the background and existing works related to the project, particularly focusing on the MERN stack and frontend technologies.

**Internship Activities**

Describes the tasks, roles, responsibilities, and weekly activities carried out during the internship.

**Conclusion and outcomes**

Summarize the internship experience, outcomes, and lessons learned.

# CHAPTER 2: INTRODUCTION TO ORGANIZATION

## 2.1. Organization Details

DanpheLink Academy and Technology is a well-established organization that bridges the gap between IT solutions and education. Its Technology division delivers expertise in Software & Web Development, Cybersecurity, and 3D Animation & VFX, serving clients across Nepal, India, and the U.A.E. Meanwhile, DanpheLink Academy offers immersive, practical training programs aimed at equipping aspiring professionals with industry-relevant technical skills. My internship provided me with firsthand experience in a dynamic workplace where technological advancements and educational excellence come together to foster digital transformation.

## 2.2. Organizational Hierarchy

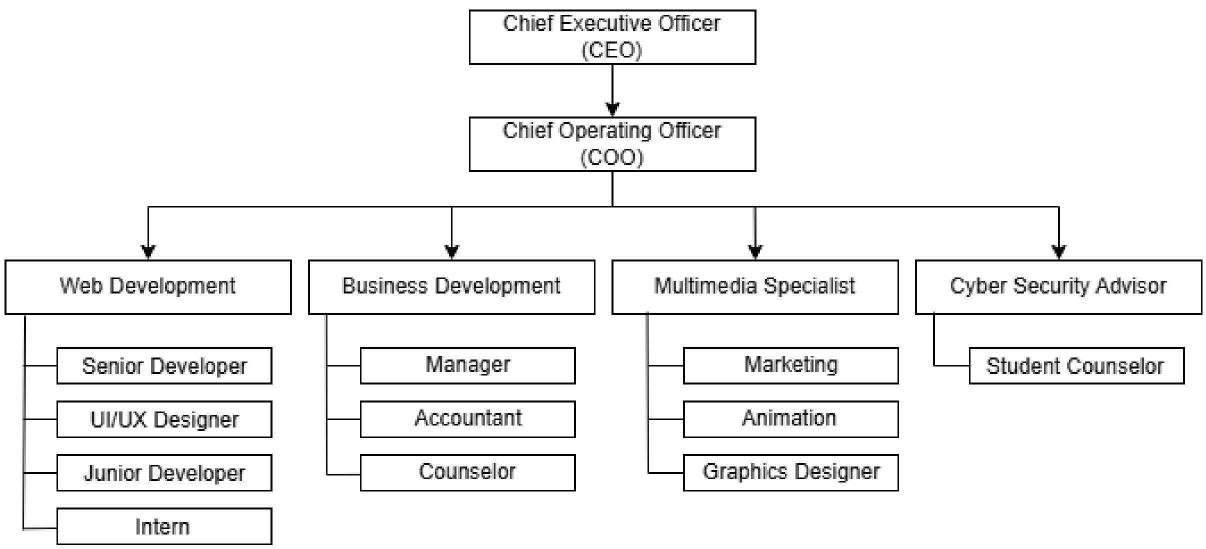


**Figure 2.2.1: Organization Hierarchy**

## 2.3. Working Domains of Organization

DanpheLink Academy and Technology operates in two main domains:

- **Technology Solutions:** Specializing in Software & Web Development, Cybersecurity, and 3D Animation & VFX, the organization delivers innovative IT solutions to clients across Nepal, India, and the U.A.E.

- **Education & Training:** Through DanpheLink Academy, it provides hands-on, industry-relevant training programs to equip individuals with cutting-edge technical skills.

## 2.4. Description of Intern Department/Unit

During my internship at DanpheLink Academy and Technology, I had the opportunity to work in the Technology Web Development department as part of a skilled team of eight members, including senior and junior developers, along with three interns, including myself. Our team was dedicated to developing innovative web solutions, with a key focus on building the DanpheLink website. This project fostered seamless collaboration between frontend and backend teams, resulting in a scalable, responsive, and secure platform that strengthened the organization's digital presence. Contributing to this initiative not only gave me hands-on experience with modern web technologies but also deepened my understanding of real-world software development workflows.

# CHAPTER 3: BACKGROUND STUDY AND LITERATURE REVIEW / RELATED WORKS

## 3.1. Background Study

The development of the DanpheLink website is grounded in key web development principles and modern technologies. The client-server architecture forms the basis of communication between the browser (client) and the server, ensuring efficient data exchange [1]. The MERN stack (MongoDB, Express.js, React.js, and Node.js) has been chosen for its ability to build scalable and interactive applications. React.js follows a component-based architecture, utilizing a virtual DOM to enhance rendering performance [2].

My primary responsibility during the internship was working on the frontend development of the DanpheLink website. The backend system and admin panel were already developed, and my role focused on implementing UI components and ensuring seamless frontend integration. Additionally, I made minor adjustments to the backend, but these changes were only related to modifying data display on the frontend, rather than developing new backend functionalities.

To facilitate a smooth user experience, the project adheres to RESTful API principles, allowing efficient data retrieval and standardized communication [3]. Furthermore, the website implements responsive web design, ensuring adaptability across multiple devices and screen sizes [4]. By integrating these core technologies, the DanpheLink website provides optimized performance, scalability, and an enhanced user experience.

## 3.2. Literature Review

The development of modern web applications relies on a combination of scalable architectures, efficient frontend frameworks, and optimized UI/UX design. This literature review examines existing research on Single Page Applications (SPAs), the MERN stack, frontend-backend integration, and performance optimization to provide a theoretical foundation for the development of the DanpheLink website.

**1. Single Page Applications (SPA) and React.js**

Single Page Applications (SPAs) have gained popularity due to their ability to provide a seamless user experience by dynamically updating content without requiring full-page reloads. Research highlights that React.js, as a frontend framework, enhances SPA performance through its **virtual DOM** mechanism, which efficiently updates only the necessary parts of the

UI rather than re-rendering the entire page [5]. Studies also emphasize that React's component-based structure improves reusability and maintainability, making it a preferred choice for modern web applications [6].

**2. Scalability in MERN Stack Applications**

The MERN stack (MongoDB, Express.js, React.js, and Node.js) is widely used for building **full-stack JavaScript applications** due to its flexibility and scalability. Research suggests that MongoDB's NoSQL structure allows efficient handling of dynamic data, while Node.js ensures non-blocking operations for better performance under high traffic [7]. Studies also highlight that MERN-based applications benefit from a unified JavaScript ecosystem, simplifying data flow between frontend and backend [8].

**3. Optimized Frontend-Backend Communication**

Effective communication between frontend and backend is crucial for ensuring smooth data flow and reducing performance bottlenecks. Studies indicate that RESTful API design enhances structured data retrieval, leading to faster response times and improved application efficiency [9]. Research also emphasizes the importance of managing state effectively in frontend applications to minimize redundant API calls and optimize user experience[10]. While backend development was not my primary responsibility, I worked on refining backend responses to improve frontend data visualization.

**4. Responsive Web Design and UI/UX Best Practices**

Ensuring a consistent user experience across multiple devices is a fundamental aspect of modern web applications. Literature on responsive web design (RWD) suggests that flexible grid layouts, media queries, and adaptive UI elements enhance accessibility and usability. Research further emphasizes that intuitive navigation, well-structured UI components, and optimized page load times contribute to a positive user experience. In my role, I applied these UI/UX principles to improve the design consistency and responsiveness of the DanpheLink website.

**5. Frontend Performance Optimization**

Frontend performance plays a key role in the efficiency of web applications. Studies highlight that minimizing unnecessary re-renders, optimizing component structure, and implementing efficient state management significantly improve application responsiveness and load times. Research also suggests that techniques such as lazy loading, code splitting, and reducing DOM manipulation contribute to better performance. These findings were particularly relevant to my work, as I focused on improving the efficiency of React components and reducing frontend performance bottlenecks.

In conclusion, The insights from these studies were directly applicable to my contributions to the DanpheLink website. By leveraging React.js for SPA architecture, applying best practices in UI/UX design, and optimizing frontend performance, I was able to enhance the usability and efficiency of the application. My internship experience provided practical exposure to modern web development concepts, reinforcing the importance of these technologies in building scalable, high-performance applications.

# CHAPTER 4: INTERNSHIP ACTIVITIES

## 4.1 Roles and Responsibilities

During my 2-month internship at DanpheLink Pvt. Ltd, my primary focus was on enhancing my frontend development skills. Although I was hired as a full-stack MERN developer, the first phase of my internship revolved around strengthening my knowledge in frontend technologies like HTML, CSS, Tailwind Css, JavaScript, and React. I also explored TypeScript, which helped improve my code quality by introducing static typing, enabling better maintainability, and streamlining debugging. This deeper understanding of frontend development allowed me to build more structured, scalable, and error-free applications.

In addition to coding, I had the opportunity to work with Figma, a powerful design tool used for creating UI/UX wireframes and prototypes. This was crucial for developing a strong understanding of UI/UX principles, as it allowed me to collaborate effectively with designers. By interpreting their design files, ensuring design consistency, and focusing on pixel-perfect layouts, I bridged the gap between design and development. This experience not only improved my technical skills but also enhanced my ability to understand and implement user-centered design.

Another key aspect of my learning was project management and collaboration. I was introduced to Trello, a task management tool used within Agile workflows, which helped me grasp how tasks are organized and tracked during sprints. By working within an Agile framework, I learned the importance of breaking down tasks into smaller, manageable units, setting achievable goals, and maintaining open communication with the team. This experience strengthened my ability to work in a collaborative environment, improving both my technical and soft skills, and has better prepared me for the full-stack development tasks I will undertake in the next phase of my internship.

## 4.2. Weekly Log

Different activities were done during the course of internship. In general, the entire task done can be shown on weekly basis as given below:

**Table 4.1: Weekly log**

| Week | Category/Task Type | Tasks Performed |
|---|---|---|
| Week 1 | User story, useCase diagram, er-diagram, flow chart | -Introduction to User story,<br>- useCase diagram, flowchart and er-diagram practices<br>- User story, useCase diagram, flowchart and er-diagram for intern website(Academy website) |
| Week 2 | Figma Design & Prototyping | - Worked on the college website Figma design<br>- Designed About us, Our team, Program listing page, and Program detail page, Admission page,<br>- Redesign of the Homepage<br>- Designed Home and About page. |
| Week 3 | UI Implementation & Technical Research | - Completed initial UI components and started research in TSX<br>- Practiced TypeScript and solved related questions<br>- Further explored TypeScript concepts<br>- Worked on form handling<br>-Introduction to Next.js<br>- Initiated development with the Aboutpage |
| Week 4 | Development of Pages & Feature Addition | - Completed the about page and Admission page<br>-Worked on Governance Structure page<br>- Updated the danphelink Academy website course section and blog section<br>- Enhanced responsive design and made content/image adjustments |
| Week 5 | UI Refinement & Code Maintenance | - Implemented changes such as course updates and added a header logo<br>- Enhanced responsiveness and performed code |

| | | |
|---|---|---|
| | | refactoring<br>- Made not found page and bug fixes<br>- Held meetings to review and plan further refactoring work |
| Week 6 | Visual & Functional Enhancements | - Fixing logo Problem<br>- updated blog and Changed favicon in intern project<br>- updated contact form and contact page<br>- Adjusted font size and color across the website<br>- fixed social media and add links in footer |
| Week 7 | New Component Development & Responsiveness | - Attend OpenHouse Meeting and discussed about danohelink' Technology website<br>- updated meta details on both websites(danphelink's academy and technology)<br>- Resolved navbar view problem in technology website<br>-added animation on mobile view<br>- make blog card component<br>- added our partner section in footer |
| Week 8 | Final UI Refinements & Feature Integration | - Added view more button in pricing details and fixed its animation while closing and opening the details<br>- updated pricing details json file in digital marketing section<br>- make our approaches section responsive for services in technology website<br>- blog detail page update<br>- worked on a homepage herosection of academy website |

## 4.3. Description of the Project involved during Internship

**Temporary Academy Website Project**

- **Overview:**

  A complete Academy website developed using TSX, aimed at providing a modern, interactive platform for Academic information and services which is use as temporary website for certain period of time.
- **Objective:**

  To redesign and build a dynamic website that improves user experience, incorporates responsive design, and delivers essential Academy-related content seamlessly.
- **Tools & Technologies:**
  - **Development:** TSX, TypeScript
  - **Design:** Figma (used for initial design concepts and wireframes)
- **Outcome:**

  A fully functional Academy website with a modern interface, enhanced navigation, and improved visual appeal that aligns with contemporary web standards.

**2. Academy Website**

- **Overview:**

  This is the main academic website of danphelink (Danphelink.com.np) .An academic portal project focused on the Academy section, which includes an administrative panel for managing content.
- **Objective:**

  To update and maintain the Academy portal by refreshing data, managing batches, pictures, updating courses, and ensuring the content is current and engaging.
- **Tools & Technologies:**
  - **Design:** Frontend design primarily executed in Figma
  - **Development:** React.js, Next.js ,node.js,express.js and mongodb
  - **Content Management:** Administrative panel
- **Outcome:**

  A streamlined administrative experience for managing academic content, ensuring that course information, batch, images, and other details are always up to date for both administrators and end users.

**3. Corporate Website**

- **Overview:**

  A corporate website for DanpheLink, built as a frontend-only project using Next.js (TSX) and styled with Tailwind CSS

- **Objective:**

  To develop a visually appealing and highly responsive corporate website with a focus on the product page, aiming to strengthen the institution's brand identity and enhance user engagement.

- **Tools & Technologies:**
    - **Development:** Next.js, TSX,
    - **Design:** Figma
    - **Styling:** Tailwind CSS

- **Outcome:**

  A modern, responsive corporate college website that delivers a polished product page experience, with additional enhancements planned to further improve interactivity and user experience.

## 4.4. Tasks/ Activities Performed

During my internship, I was actively involved in various technical tasks and activities, contributing to both design and development aspects of multiple projects.

**Research & Skill Development**:

I dedicated time to enhancing my technical skills by researching and practicing advanced TypeScript and TSX concepts. This included improving my understanding of form handling, component-based architecture, and other modern front-end development techniques.

• **Design**:

I created detailed Figma designs for key sections, including the Homepage, About Us, Our Teams, Events, About, and admission page, notfound page ensuring a modern and user-friendly interface. Additionally, I worked on refining the UI for the Corporate College and Academy projects, focusing on consistency and usability.

• **Project Management & Communication**:

To streamline workflows and ensure effective collaboration, I utilized Trello for task management, organizing tasks, and tracking project progress. Communication with the team and stakeholders was facilitated through Slack, enabling seamless coordination.

• **UI Enhancements & Bug Fixes**:

Throughout the internship, I focused on improving user experience by implementing responsive design adjustments, refining visual elements (such as hover effects, color schemes, and card layouts), and ensuring overall consistency. Additionally, I performed code refactoring to maintain a clean and efficient codebase while resolving issues like duplication errors and misaligned components.

• **Front-end Development**:

I developed the College Website using TSX and TypeScript, implementing responsive layouts and interactive UI components. For the Corporate College website, I leveraged Next.js, TSX, and Tailwind CSS to create a visually appealing and fully responsive product page. In the Academy portal, I updated and managed front-end content, including batch images and course details, via the administrative panel.

Overall, my internship provided a well-rounded experience, allowing me to contribute to multiple projects while honing my skills in design, front-end development, and project management.

# CHAPTER 5: CONCLUSION AND LEARNING OUTCOMES

## 5.1. Conclusions

My internship provided a fantastic opportunity to apply both my technical and design skills in a real-world context, contributing to the development and enhancement of various web projects. Throughout the experience, I gained hands-on exposure to front-end development, UI/UX design, and project management, allowing me to bridge the gap between theoretical knowledge and practical implementation.A key takeaway from my internship was the value of collaborative development and continuous learning. By working on projects such as the College Website, Academy, and Corporate College, I expanded my understanding of TypeScript, TSX, Next.js, and Tailwind CSS, while further developing my skills in responsive design, UI improvements, and component-based architecture. Using tools like Figma, Trello, and Slack also helped me stay organized and maintain effective communication within a team environment.

This experience highlighted the importance of code optimization, accessibility, and user experience in web development. From writing clean and maintainable code to improving website responsiveness, every task helped me grow as a front-end developer.

Looking ahead, I am excited to continue exploring advanced web development frameworks, design principles, and full-stack development practices. This internship has not only strengthened my technical skills but also laid a solid foundation for ongoing learning and professional growth. As technology evolves, I am eager to face new challenges and expand my capabilities to build even more engaging, scalable, and user-friendly web applications.

## 5.2. Learning Outcomes

During my internship, I gained practical experience in web design, front-end development, and project management, allowing me to apply my technical skills in a professional setting. I enhanced my proficiency in TypeScript, TSX, and Next.js, developing scalable and interactive web applications while improving my expertise in Tailwind **CSS** for building responsive and visually appealing interfaces. Additionally, I worked extensively on UI/UX design using Figma, creating intuitive layouts and ensuring accessibility across different devices through responsive design principles.

A key focus of my role involved code optimization and best practices, where I applied refactoring techniques to improve efficiency and maintainability. I debugged and resolved issues related to duplication, styling inconsistencies, and component misalignment, ensuring a seamless user experience. My exposure to component-based architecture and state management helped me build reusable and scalable UI components, aligning with modern development standards.

Beyond technical development, I gained valuable experience in project management and team collaboration. I effectively managed tasks using Trello, enabling better organization and progress tracking, while communication via Slack allowed for smooth coordination with team members and stakeholders. Managing multiple projects simultaneously helped me develop strong time management skills, enabling me to prioritize tasks and meet deadlines efficiently.

This internship also contributed to my professional and personal growth by strengthening my problem-solving abilities and adaptability in a structured development environment. I learned to incorporate feedback, iterate on designs and code, and align with project requirements while following industry best practices. Overall, this experience has provided me with a solid foundation in front-end web development and UI/UX design, preparing me for more advanced challenges in the field. Moving forward, I aim to expand my expertise by staying updated with emerging technologies and exploring full-stack development opportunities.

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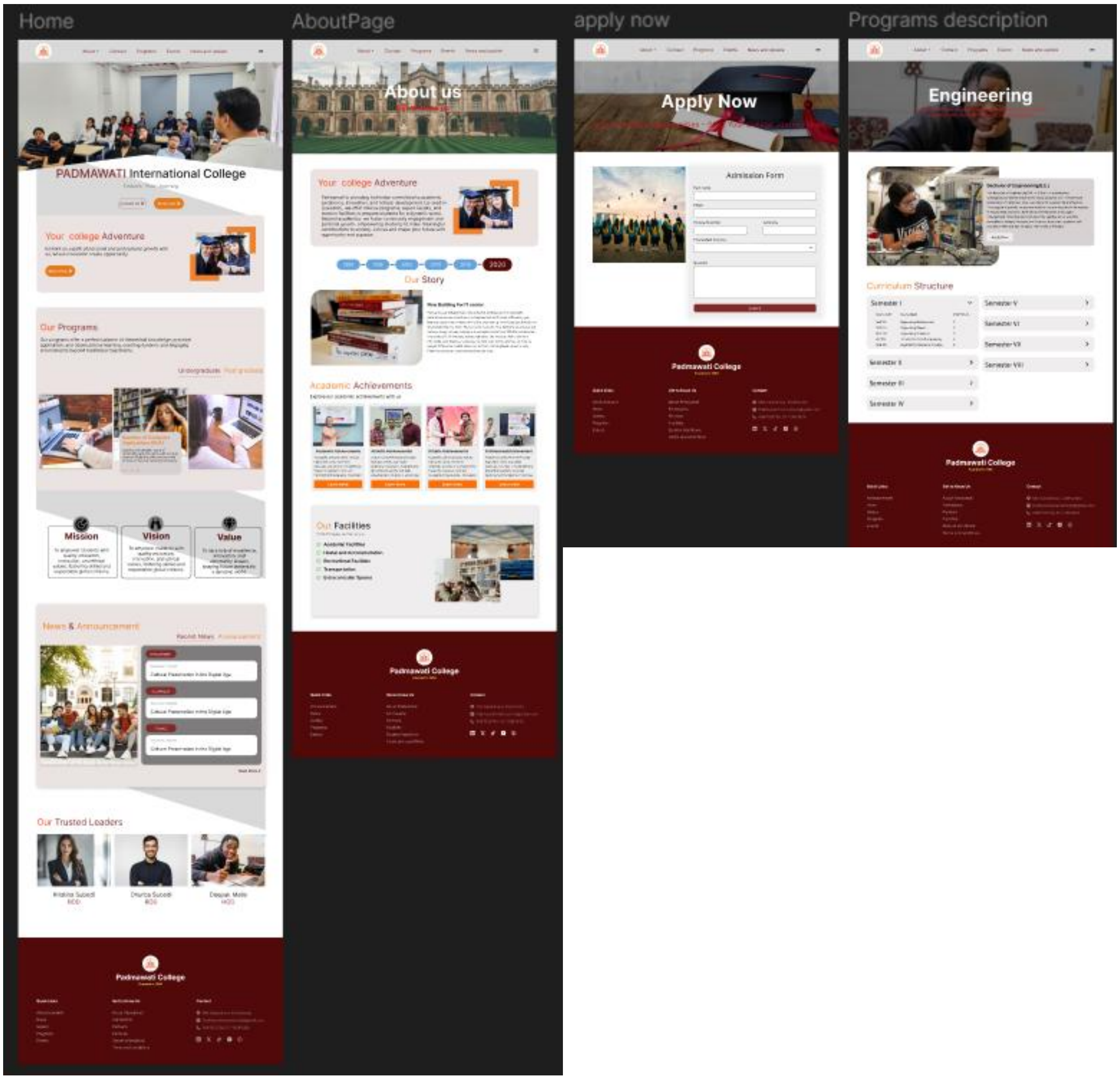
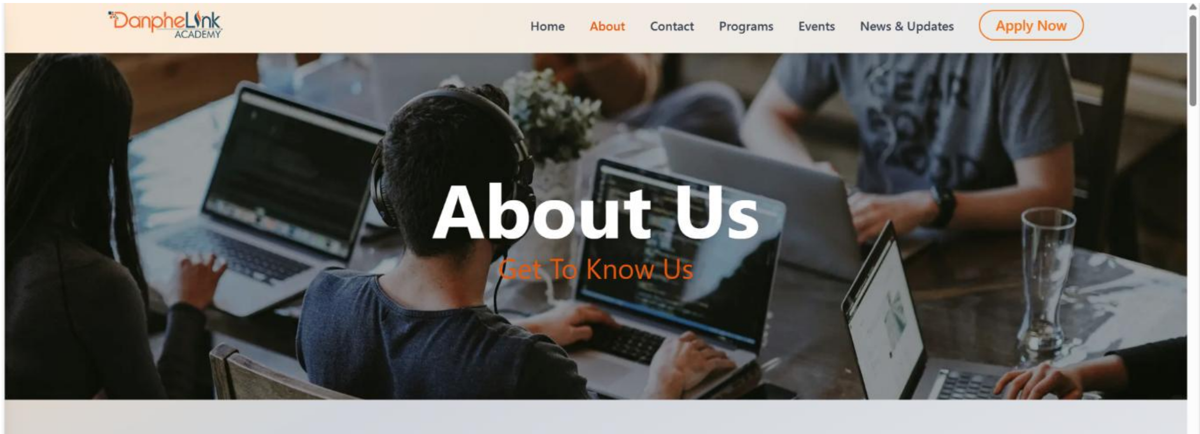
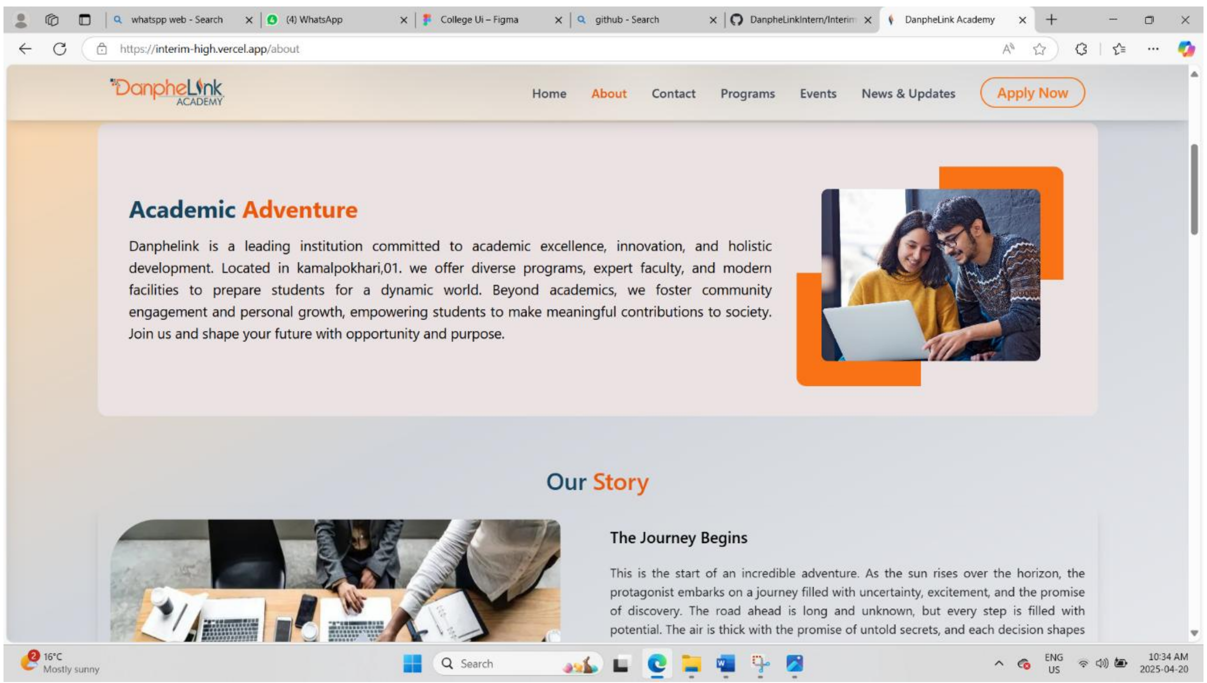
# APPENDIX

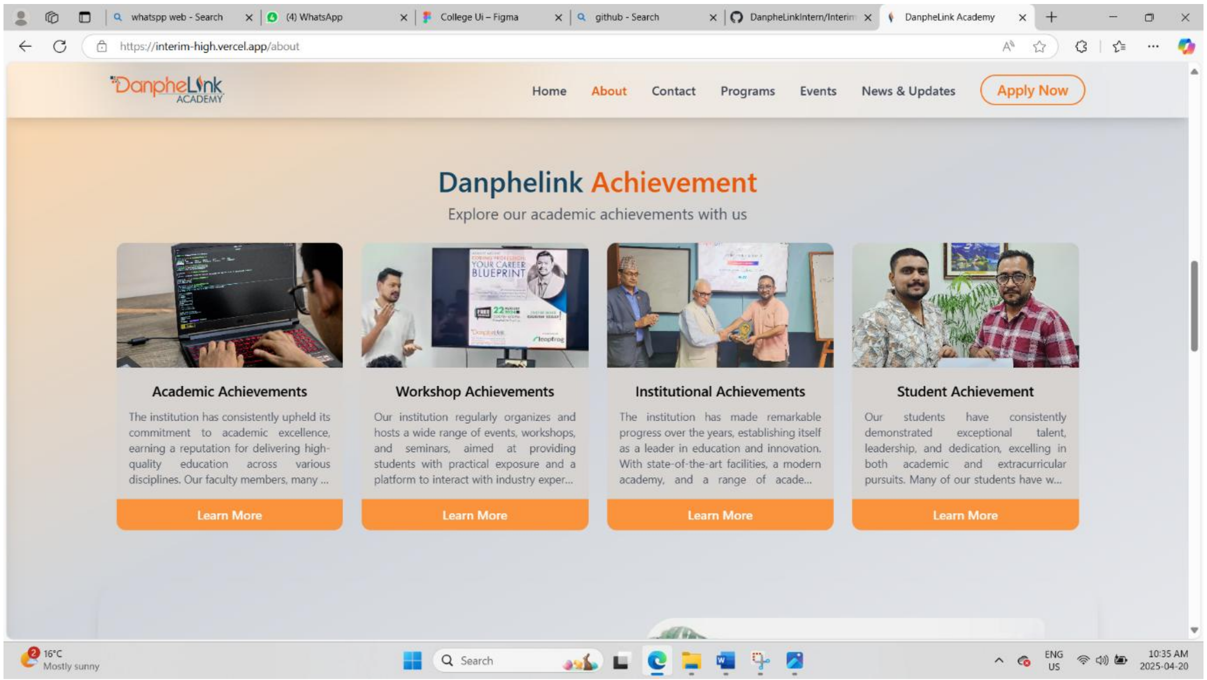
## UI/UX Design



## Temporary Academy Website Project

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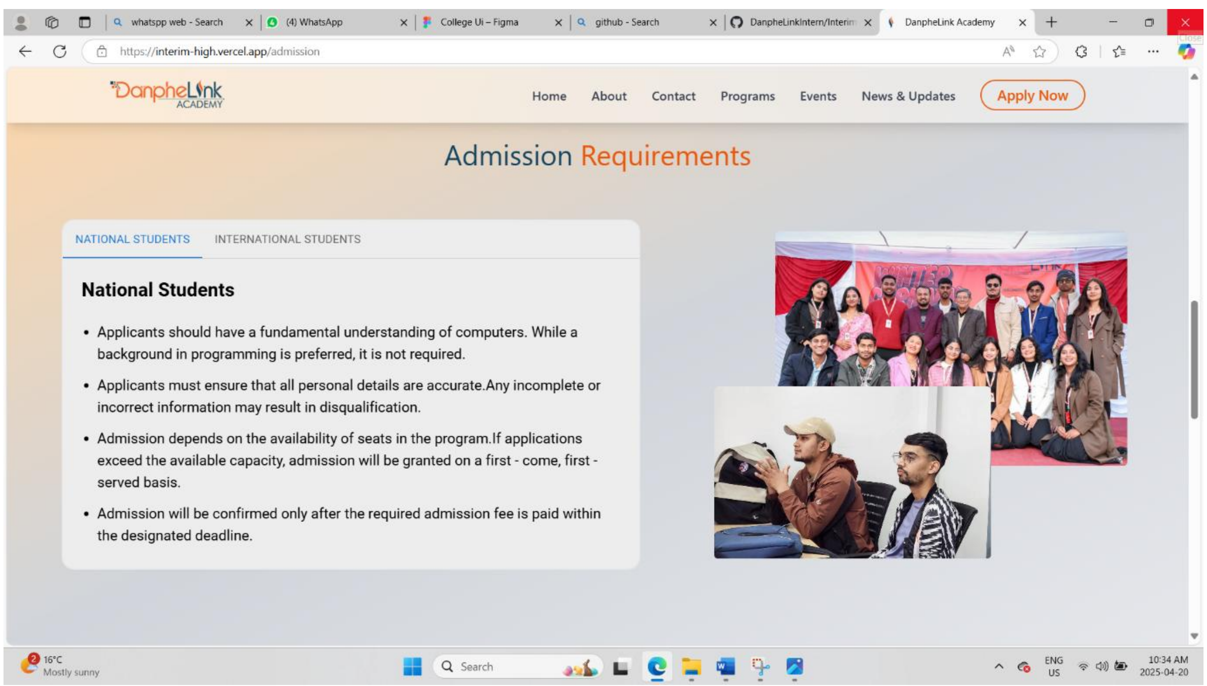
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Admission Requirements
NATIONAL STUDENTS
INTERNATIONAL STUDENTS
National Students
• Applicants should have a fundamental understanding of computers. While a background in programming is preferred, it is not required.
• Applicants must ensure that all personal details are accurate.Any incomplete or incorrect information may result in disqualification.
• Admission depends on the availability of seats in the program.If applications exceed the available capacity, admission will be granted on a first - come, first - served basis.
• Admission will be confirmed only after the required admission fee is paid within the designated deadline.
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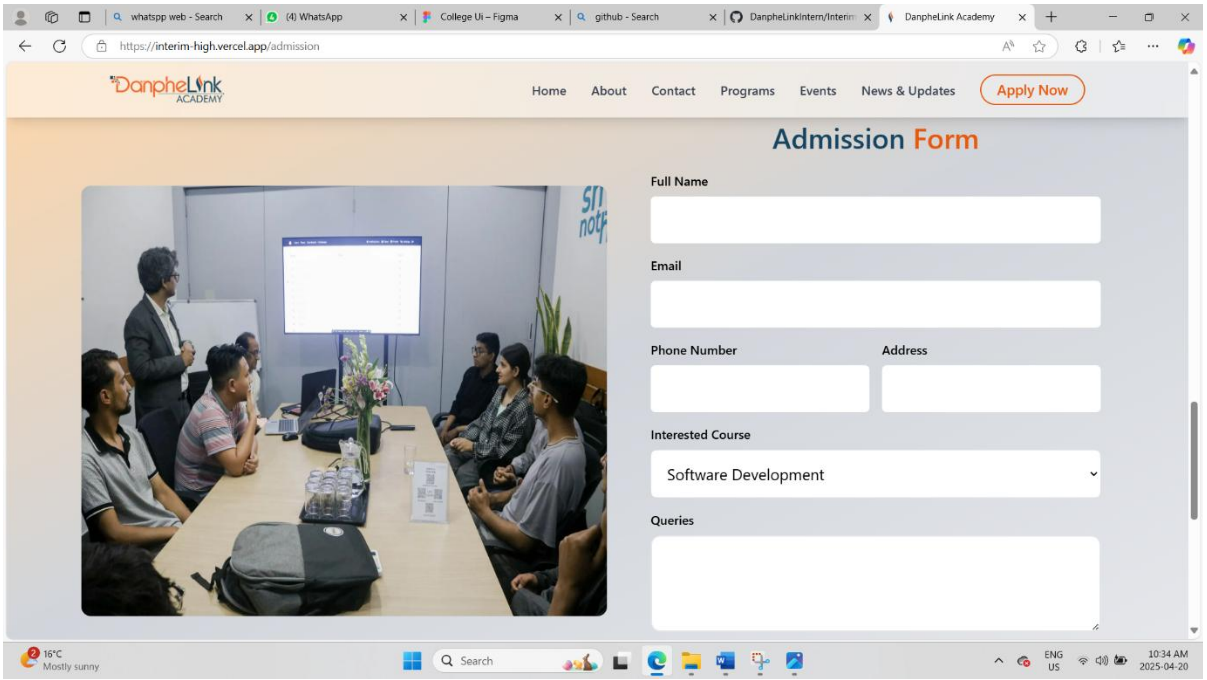
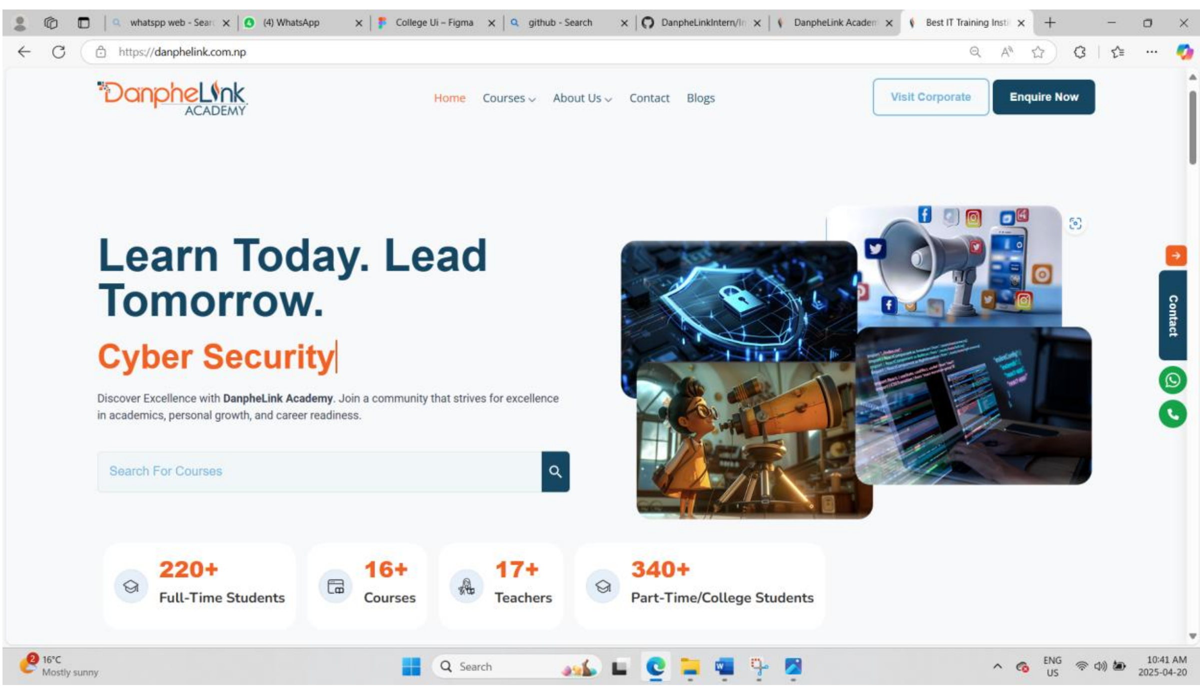
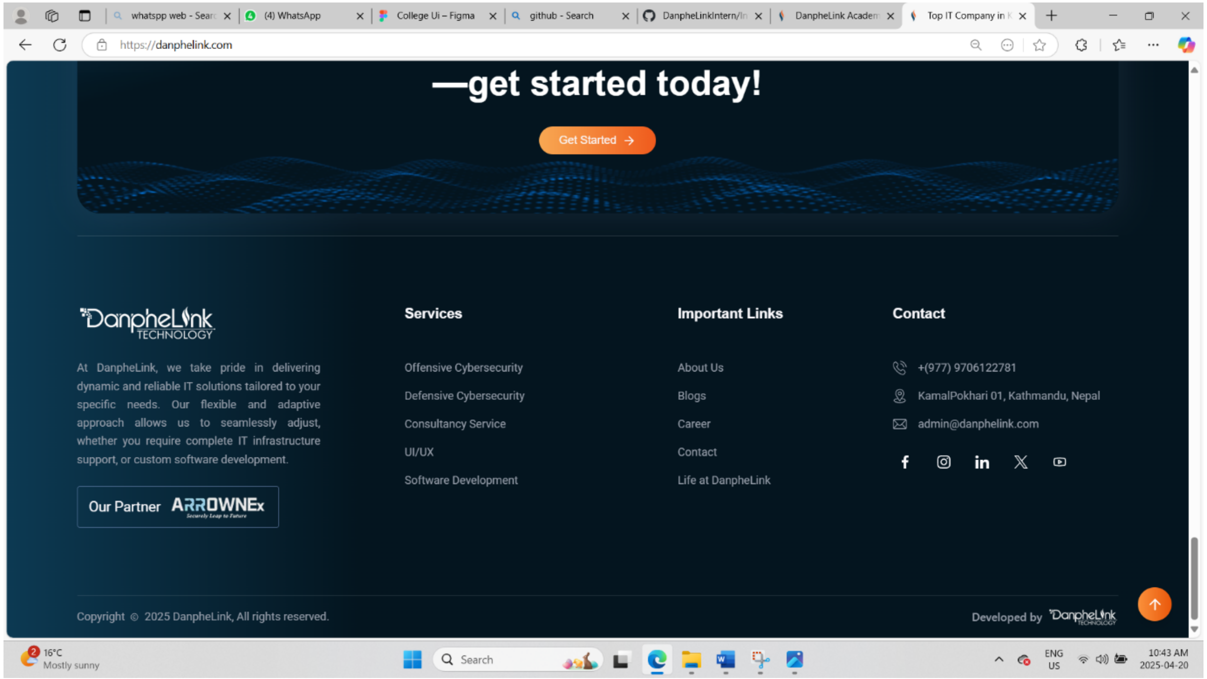
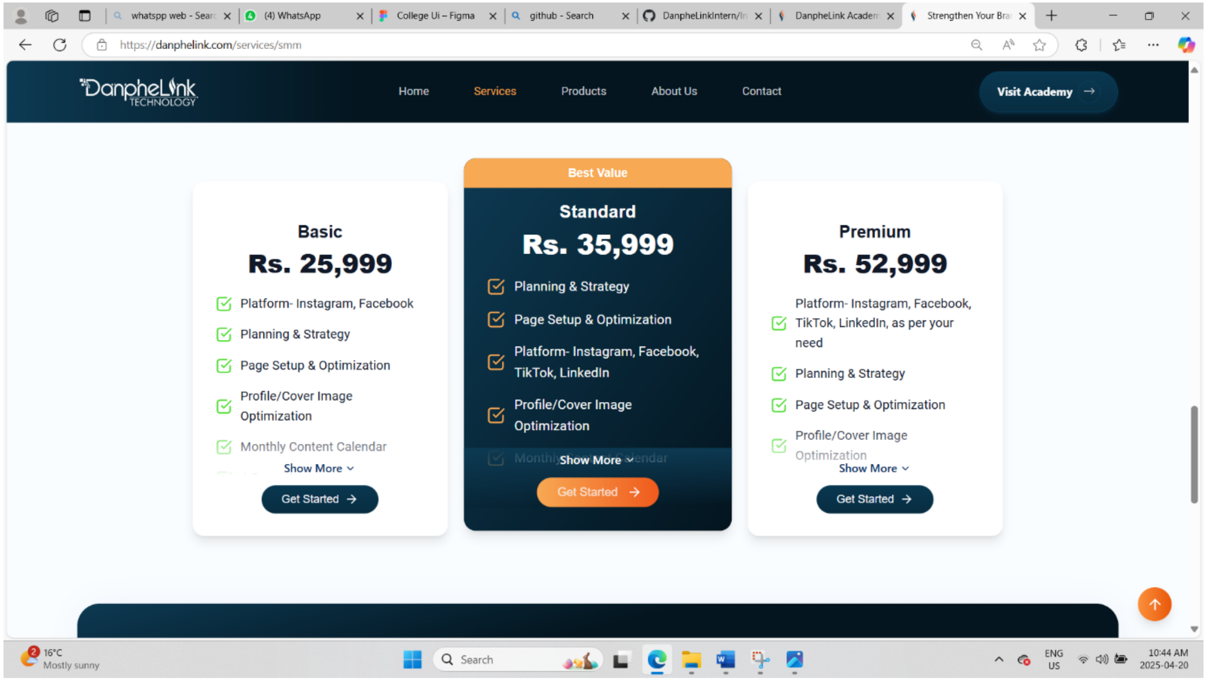



## Academy Website Project



# Technology Website

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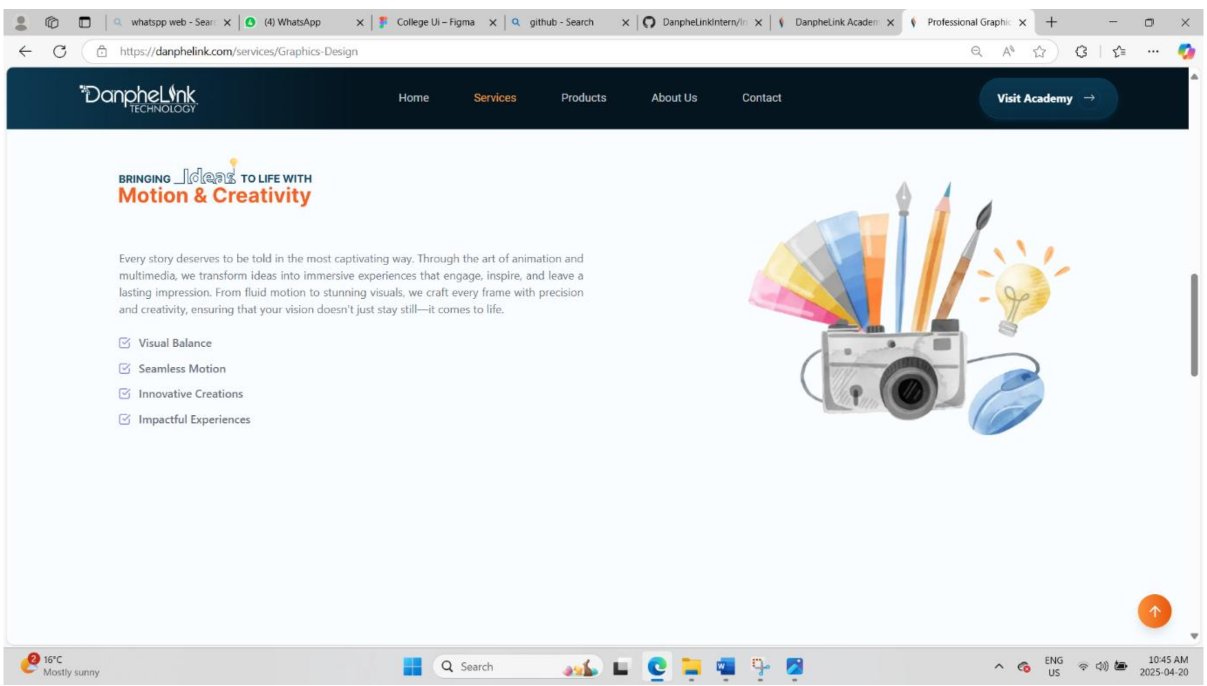
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